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Understanding the decisions of CNNs: An in-model approach

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ABSTRACT

With the outstanding predictive performance of Convolutional Neural Networks on different tasks and their widespread use in real-world scenarios, it is essential to understand and trust these black-box models. While most of the literature focuses on post-model methods, we propose a novel in-model joint architecture, composed by an explainer and a classifier. This architecture outputs not only a class label, but also a visual explanation of such decision, without the need for additional labelled data to train the explainer besides the image class. The model is trained end-to-end, with the classifier taking as input an image and the explainer's resulting explanation, thus allowing for the classifier to focus on the relevant areas of such explanation. Moreover, this approach can be employed with any classifier, provided that the necessary connections to the explainer are made. We also propose a three-phase training process and two alternative custom loss functions that regularise the produced explanations and encourage desired properties, such as sparsity and spatial contiguity. The architecture was validated in two datasets (a subset of ImageNet and a cervical cancer dataset) and the obtained results show that it is able to produce meaningful image- and class-dependent visual explanations, without direct supervision, aligned with intuitive visual features associated with the data. Quantitative assessment of explanation quality was conducted through iterative perturbation of the input image according to the explanation heatmaps. The impact on classification performance is studied in terms of average function value and AOPC (Area Over the MoRF (Most Relevant First) Curve). For further evaluation, we propose POMPOM (Percentage of Meaningful Pixels Outside the Mask) as another measurable criteria of explanation goodness. These analyses showed that the proposed method outperformed state-of-the-art post-model methods, such as LRP (Layer-wise Relevance Propagation).

1. Introduction

The emergence of Deep Learning (DL) changed the Machine Learning (ML) paradigm in recent years, especially in Computer Vision (CV). Due to their significant performance improvement in tasks like classification, these models became dominant. They have since been applied to tackle different CV problems - e.g. detection, segmentation, depth estimation.

Despite this overwhelming ubiquitousness, DL models and in particular Convolutional Neural Networks (CNNs), are for the most part considered black-box models, for which it is hard to understand the reasons for the labels that they generate. These models perform millions of non-linear operations, rendering it impossible for humans to follow this enormous amount of computations. Furthermore, contrarily to what happened with traditional ML methods, the whole process is done end-to-end, from raw input to output, implicitly transforming the input into the most convenient feature space for the model's calculations, eliminating the need for feature engineering. This automatic feature extraction, although highly advantageous, adds an extra layer of complexity and opaqueness to these models. In fact, there is an important trade-off between predictive performance and opaqueness.

Therefore, understanding the decisions of DL models in general is of the utmost importance to allow the deployment of robust, transparent and trustworthy DL-based systems in real-world scenarios. This is particularly relevant in critical high-stakes areas, where the consequences of a wrong decision could greatly affect its users, or even lead to the loss of human lives, e.g. medicine, criminal justice, and autonomous driving. For these reasons, there is an increasing interest in finding solutions to tackle, or, at least, mitigate this limitation, and that can comply with recent legislation introduced in the European Union (EU) about the right to algorithmic explanation.

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However, this is still a relatively new research area, that lacks unified definitions of core concepts, such as interpretability and explainability, as well as a universally accepted taxonomy and quantitative evaluation frameworks. The terms explainability and interpretability are often used interchangeably in the literature. In this work, we adopt the definition proposed by Gilpin et al. [7]. The authors loosely define interpretability as the process of understanding the model's internals and describing them in a way that is understandable to humans. Thus, an explainable model is one that can summarise the reasons for its behaviour. In fact, for a model to be explainable, it needs to be interpretable, but also complete, i.e., to describe the system's internals accurately. As such, explainable models are interpretable by default, while the reverse may not be true. Therefore, a good explanation should be able to balance the interpretability-completeness trade-off.

Producing meaningful explanations is an user- and domain-dependent task. Moreover, it is imperative that new solutions are able to generate explanations in an unsupervised fashion, without needing additionally labelled data. In order to mitigate this need, we extend on the previously proposed explainer + classifier joint approach [16], by further detailing the rationale behind its architecture. We also propose a custom three phase training process, along with two loss functions. Furthermore, the proposed architecture is quantitatively and qualitatively tested on real data, and an evaluation metric is proposed.

The remainder of this article is structured as follows. Section 2 discusses related work, while Section 3 presents the overall architecture of the proposed in-model approach, and its training process. Section 4 details the datasets used to evaluate the method and how the hyperparameters were defined. Results and evaluation metrics are presented and discussed in Section 5. Finally, Section 6 summarises the main conclusions and future work.

2. Related work

The majority of the literature focuses only on interpretability, and more specifically on post-model approaches, i.e., methods that are applied after the model is trained. These methods can be model-agnostic, such as LIME (Local Interpretable Model-agnostic Explanations), which locally approximates any model by a simpler and more interpretable one [15].

However, most methods are specific to neural networks and fall under the "gradient-based" category, in which pixel-wise relevance is backwards propagated through the network until the input space is reached; the produced explanations are heatmaps highlighting the most relevant pixels for the classification decision. Examples include: Input $\times$ Gradient (element-wise multiplication between input and gradient of the output layer with respect to the input); Integrated Gradients (integration of the gradient along a path from the input to a reference point - [23]); Saliency (computes a class-specific saliency map, by ranking the pixels based on their influence on the classification score - [20]); Deconvolution (application of another CNN to the output of the original classifier, built to undo the operations performed by the latter - [24]); Guided Backpropagation (extension of Deconvolution that sets negative gradients to zero when backpropagating through ReLU units - [22]); Guided Grad-CAM (Grad-CAM produces explanations by computing the gradient of the class score (logit) with respect to the feature map of the last convolutional layer of the network. Guided Grad-CAM results from the combination of Grad-CAM with Guided Backpropagation through an element-wise product, to achieve pixel-level granularity. - [19]); LRP (redistributes positive and negative relevance recursively to each neuron's input, according to its contribution to the neuron's output, while enforcing relevance conservation - [2]). Although most of these methods produce meaningful explanations, Deconvolution explanations are usually noisier and resemble filter visualisations, and Guided Grad-CAM and Guided Backpropagation were disproved by Adebayo et al. [1].

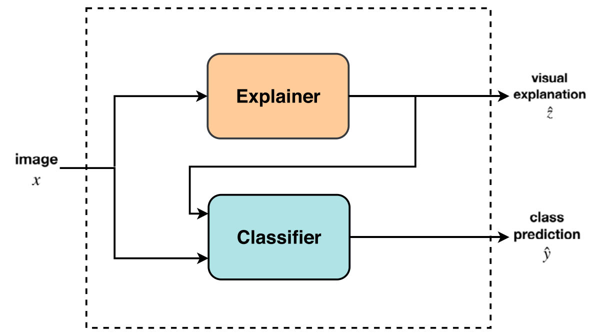


Fig. 1. Block diagram of the proposed architecture, composed by an explainer and a classifier, trained end-to-end through a joint loss function. It outputs the class label, $\hat{y}$, as well as the corresponding visual explanation, $\hat{z}$.

A different research line pertains to producing example-based explanations: selecting certain instances as the explanation. This category includes methods such as prototype and criticism generation [13], and complementary examples [11]; this recent work constitutes one of the few in-model explainability methods. However, this method produces complemental example- and text-based explanations, while ours produces visual explanations.

3. Proposed joint architecture

3.1. Overview

We propose an end-to-end in-model joint approach, based on an explainer+classifier architecture. A simple block diagram of the model can be found in Fig. 1 and its concretisation is depicted in Fig. 2.

This architecture outputs a class prediction, $\hat{y}$, and what we call a visual explanation of such decision, $\hat{z}$. The explainer takes as input an image, x, and outputs the corresponding visual explanation with the same spatial dimensions as the input. On the other hand, the classifier not only takes as input the same image, but also the output of the explainer, i.e., it is trained using the explanation. Therefore, the classifier focuses only on the image regions the explainer deemed relevant and does not take into account regions where the explanation for that class is not present. This approach aligns with the intuition that, when explaining a decision, for example, whether or not an image contains a car, humans tend to first separate what is an object and what is not, and then proceed to look for patterns in the region where the object is in order to classify it correctly. Conversely, sometimes humans cannot tell if an object belongs to some class, but can tell which regions of the image do not contain said class.

3.2. Classifier

The initial classifier (bottom row in Fig. 2) was inspired by the VGG architecture proposed by Simonyan and Zisserman [21], having 4 consecutive convolution-convolution-pooling stages, ending with 3 fully connected layers and a softmax layer. Convolutional layers use 3×3 kernels and ReLU activation functions, while pooling layers use max pooling with a stride of 2. The number of filters and the spatial dimensions decrease as a power of 2 with the stage level, starting at 32 and 224×224, respectively. Pooling layers take as input the result of a multiplication layer (purple connections and layers in Fig. 2). This layer performs the element-wise multiplication of the preceding classifier layer's output with the correctly downsampled version of the explainer's output by average

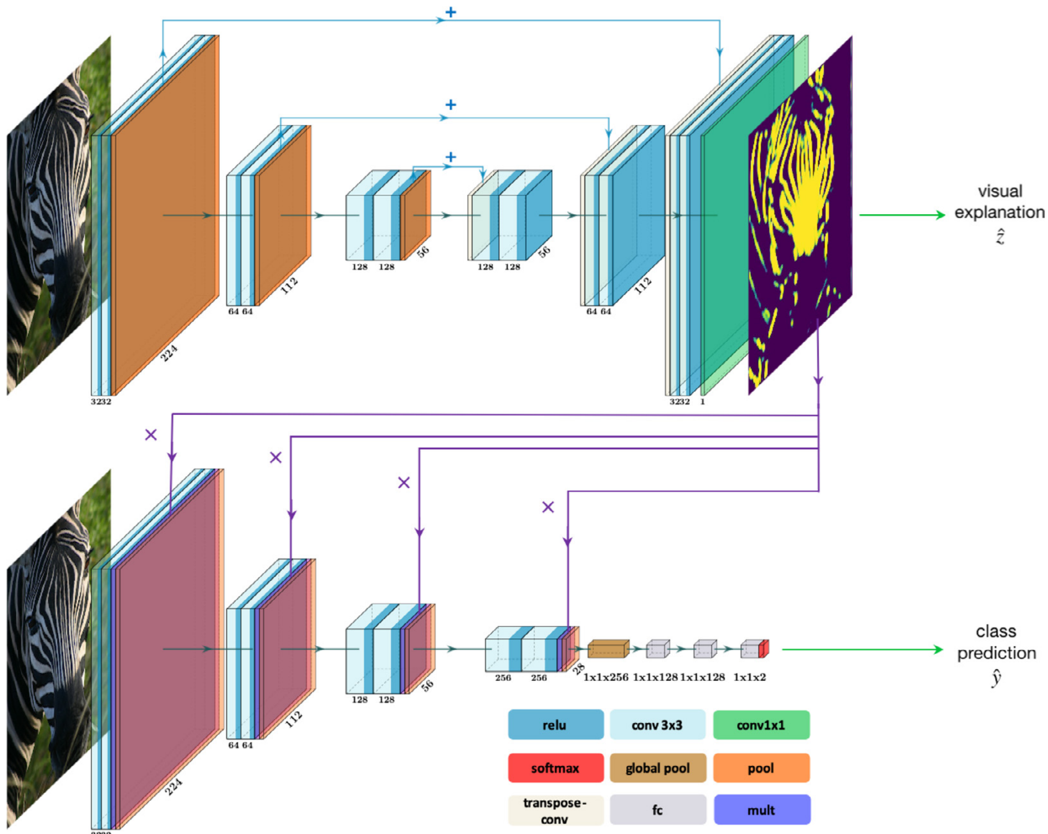


Fig. 2. Detailed view of the proposed classifier+explainer architecture. The explainer (top row) is based on an encoder-decoder network and the classifier (bottom row) is based on VGG-16. The sums correspond to the simple element-wise addition of the outputs of the respective layers. The multiplications involve the concatenation and resizing of the explainers output before computing the element-wise multiplication with the involved classifier layer. Although our first implementation involved this modified VGG-16 architecture, any CNN classifier can be used, provided the necessary connections are correctly made and the whole model is trained end-to-end. This diagram was built using https://github.com/HarisIqbal88/PlotNeuralNet.git. (Best viewed in colour.)

pooling.[1] These connections allow the classifier to focus only on the important image regions highlighted by the explainer. A global pooling layer succeeds the 4 convolutional stages and the network ends in two $1 \times 1 \times 128$ fully connected layers with sigmoid activation functions and 30% dropout rate, followed by a fully connected layer with a softmax activation.

However, any classifier can be used, provided that the correct connections are introduced. In fact, the performed experiments (refer to Section 5) include the ResNet-18 [9] as a classifier, modified only to introduce the aforementioned connections. As the proposed architecture aims at explaining the classifier's decisions, the classifier should be chosen first, depending on the classification problem and the available data; the explainer must adapt to the classifier and not the other way around.

3.3. Explainer

The explainer (top row in Fig. 2) consists of a convolutional encoder-decoder network, based on U-Net proposed by Ronneberger et al. [17]. It outputs a visual explanation, $\hat{z}$.

The downsampling path, or encoder, is a simple convolution-convolution-pooling scheme repeated 3 times, similar to the VGG-based classifier. Each convolutional layer uses 3×3 kernels and is followed by a ReLU activation layer. All pooling layers use max pooling and a downsampling factor of 2. The first convolution-convolution-pooling stage has 32 stacked filters and the following layers increase the number of filters by a power of 2.

The upsampling path, or decoder, follows a "transpose convolution-convolution-convolution" scheme, where the first convolution operation is applied to the pixel-wise sum of the previous layers output with the corresponding convolutional layer in the downsampling path. The whole decoder is comprised of 3 of these stages, ending in a convolutional layer with a 1×1 kernel, used to reduce the filter space dimensionality from 32 to 1. This last convolutional layer has a linear activation and is followed by a batch normalisation layer, an hyperbolic tangent activation layer and a ReLU activation layer. Conversely to what is done in the downsampling path, the number of filters starts at 128 and decreases as a power of 2.

3.4. Training

Training this architecture involves three phases. First, in **phase 1**, only the classifier is trained, while the explainer remains frozen. It is worth noting that the explainer is initialised in such a way that the initial explanations consist of white images, i.e., matrices filled with ones; this initialisation is achieved by imposing a large bias (>1) in the explainer's batch normalisation layer. This means that the multiplications introduced by the explainer - purple ar-

[1] It is important that the explanations are downsampled via average pooling instead of max pooling. We verified that, through max pooling, explanations would remain at their initial state (white images - refer to Section 3.4), because the explainer would circumvent the sparsity constraint (refer to Section 3.5.1), by lowering every pixel in a receptive field except one. This way the sparsity constraint would take effect, but the classifier would still get its feature maps multiplied by white explanations, and thus remain unchanged. Through average pooling this behaviour is avoided, because the sparsity constraint can only be enforced if, on average, all the pixels in the receptive field are lowered from one.

rows in Fig. 2 - are bypassed and the classifier is not taking the explainer's output into account, as the explainer is not trained yet. The intuition is that we start by considering the whole image as an explanation and gradually eliminate irrelevant regions.

Afterwards, in **phase 2**, the process is reversed. Although the classifier remains frozen, the explainer learns by altering its explanations and assessing the corresponding impact on the classification, through the joint classification and explanation loss (refer to Section 3.5).

Finally, in **phase 3**, the whole architecture is fine-tuned end-to-end. The classifier is learning with the new information provided by the explainer, through the multiplications of the explainer's output at every classifier's stage.

It is imperative that at the end of phase 1 the classifier remains somewhat unstable, i.e., that its loss does not *plateau*, so that in phase 3 both modules can learn from each other. Otherwise, in phase 3 the classifier would not update its parameters with the new information provided by the now trained explainer and vice-versa. Therefore, in the end, both explainer and classifier improve, fruit of this dynamic interaction between the two and their respective loss functions.

3.5. Loss functions

The whole training process described above resorts to the end-to-end loss function defined in Eq. (1). This loss constitutes a weighted sum of the influence of both classifier and explainer ($\mathcal{L}_{class}$ and $\mathcal{L}_{expl}$, respectively), as realised by the α hyperparameter.

$$\mathcal{L} = \alpha\mathcal{L}_{class} + (1-\alpha)\mathcal{L}_{expl} \tag{1}$$

In general, during phases 1 and 3, $\alpha > 0.5$, because the correct classification is an indispensable part of the system; it directly affects the classifier and indirectly affects the explainer (through this joint loss, the explainer is always influenced by the classifier). However, a slight decrease in classification accuracy is tolerable, as long as it means that the system is also able to provide meaningful explanations. Conversely, during phase 2, $\alpha < 0.5$, so as to give more strength to the explainer.

The classification loss, $\mathcal{L}_{class}$, is the categorical cross-entropy loss, employed in the majority of multiclass classification scenarios. Regarding the explanation loss, two different approaches are proposed, an unsupervised and a hybrid loss.

3.5.1. Unsupervised explanation loss

As it would be very hard to supervise the explainer's training, not only because explanations are user-, context- and domain-dependent, but also because an explanation constitutes a higher conceptual level, the explainer is trained with an unsupervised loss. This loss, represented in Eq. (2), is also a weighted sum of relevant properties one wants to promote on the resulting explanations, as realised by the β hyperparameter. The properties of sparsity and contiguity are imposed by means of two regularisation terms: the penalised $l1$ norm and total variation ($\mathcal{L}_{sparsity}$ and $\mathcal{L}_{contiguity}$, respectively) [5].

$$\mathcal{L}_{expl\_unsup} = \beta\sum_{i=1}^{N}\mathcal{L}_{sparsity}(\hat{z}_i) + (1-\beta)\sum_{i=1}^{N}\mathcal{L}_{contiguity}(\hat{z}_i) \tag{2}$$

$$\mathcal{L}_{sparsity}(\hat{z}) = \frac{1}{m\times n}\sum_{i,j}|\hat{z}_{i,j}| \tag{3}$$

$$\mathcal{L}_{contiguity}(\hat{z}) = \frac{1}{m\times n}\sum_{i,j}|\hat{z}_{i+1,j}-\hat{z}_{i,j}| + |\hat{z}_{i,j+1}-\hat{z}_{i,j}| \tag{4}$$

where m and n are the spatial dimensions of feature map $\hat{z}$.

The penalised $l1$ norm (Eq. (3)) promotes sparsity, as it shrinks less important features to zero, performing feature selection. This penalty works as an explanation budget, limiting the percentage of the input image that can be considered an explanation. The greater the penalty, the more regularisation, and therefore the less explanation budget is available.

The total variation factor (Eq. (4)) promotes spatial contiguity, i.e., it encourages smoothness and spatial localisation of the activations of $\hat{z}$, by minimising its local spatial transitions. This factor was introduced because initial results hinted at the need for not only sparse, but also connected explanations, i.e., explanations in which semantically related parts of the images are connected [16].

3.5.2. Hybrid explanation loss

Imposing stronger constraints on the produced explanations without needing to fully supervise their generation is possible through a hybrid strategy, provided that some type of annotated data is available from another task. In this case, masks from an object detection task were used. The aim is to drive the explanations not to focus on regions outside the interest regions, while still encouraging sparsity and contiguity. In the case of object detection annotations, the regions of interest are the areas inside the bounding boxes. Therefore, the hybrid explanation loss is a combination of the aforementioned unsupervised loss and a weakly supervised constraint, as realised by the γ hyperparameter, as defined below:

$$\mathcal{L}_{expl\_hybrid} = \mathcal{L}_{expl\_unsup} + \gamma\left|\frac{\sum_{i,j}(1-z_{i,j})\hat{z}_{i,j}}{\sum_{i,j}(1-z_{i,j})}\right| \tag{5}$$

where $\hat{z}_{i,j}$ ($z_{i,j}$) is the value of pixel (i, j) of feature map $\hat{z}$ (mask z). This mask is an image where regions of interest are represented by ones, and the rest by zeros.

This added constraint punishes explanations outside the region of interest, by computing the product between the inverted mask ($1-z_{i,j}$) and the explanation ($\hat{z}_{i,j}$): in the region of interest, the multiplication by zeros does not punish the explanation; outside the region of interest, the explanation is multiplied by ones, so, in order to decrease the explanation loss, the explainer will have to punish these pixels (lower their value). The divisor minimises the impact of the size of the regions of interest in the mask.

4. Experimental methodology

4.1. Datasets

The proposed architecture was tested on two datasets, as detailed below.

4.1.1. ImageNetHVZ

We started by testing the proposed architecture in a dataset we call ImageNetHVZ. As the name implies, this dataset is a subset of ImageNet [4], composed only by horses and zebras (synsets n02389026 and n02391049, respectively). Of the 2600 images, we kept only the ones for which bounding boxes were available, totalling 324 images of horses and 345 images of zebras.

Data was split into 85%-15% for training and testing, giving a total of 100 images for the latter. The training set was further split into 80%-20% for training and validation.

4.1.2. NIH-NCI Cervical cancer dataset

The proposed approach was also tested on a medical dataset, the National Institute of Health/National Cancer Institute Cervical Cancer Dataset (NIH-NCI), that was collected in the Guanacaste project [10] and is publicly available.

Cervical cancer remains a significant cause of mortality in low-income countries [12] and is characterised by the abnormal growth of cells on the cervix - Cervical Intraepithelial Neoplasia (CIN). CIN can be detected by means of a colposcopy, which allows the detection of macroscopic changes in the tissue, such as colour and morphology. In this database, images are annotated with their CIN progression level: no histology (1207 images), normal (330 images), $<$ CIN2 (184 images), CIN2 (146 images), CIN3 (234 images), and cancer (19 images). Bounding boxes for the cervix regions are also available for the 2120 images. Since the data is unbalanced, a coarser division was made: instances with no histology[2] done, normal or less than CIN2 are considered as not cancerous; the rest are considered cancerous.

This dataset was split into 95%-5% for training and testing (106 images for testing), and the training set was further divided into 80%-20% for training and validation.

4.2. Data preprocessing

Preprocessing involved normalising each input image to a pixel value range of [0, 1] and performing data augmentation, specifically the “strong augmentation” suggested by the authors of Albumentations [3] in their github repository,[3] which includes random rotations, flips, blur, distortion, among others. This augmentation was performed with 0.2 probability.

4.3. Hyperparameter tuning

The proposed architecture was trained using the Stochastic Gradient Descent (SGD) optimiser. The learning rate value was chosen by first ensuring that the classifier alone would converge to a minimum. Then, the whole architecture was tested with that same learning rate to ensure that both classifier and explainer reached acceptable results. After such analysis, we chose 0.01 as the learning rate value for both modules, in all training phases. The learning rate was reduced by a factor of 0.2 every 10 epochs, during which the validation loss had *plateaued*, if a minimum of 0.00001 had not been reached. A batch size of 8 was used.

Regarding the α and β hyperparameters, α was first fine tuned, keeping β constant at 1. First, a coarse grained search was performed: {0.5, 0.75, 0.9}. Then, a finer search was conducted between the range that produced the most reasonable explanations (closer to 0.9), while maintaining the predictive performance. The β hyperparameter was tuned by searching within a broader range, [0.0, 1.0], and afterwards within a smaller set, {0.75, 0.8, 0.9, 0.99}. Finally, γ was tuned by searching in $\{10.0, 1.0, 1e-2, 1e-4\}$, while keeping the best β value.

5. Results and discussion

This section is dedicated to a thorough qualitative and quantitative analysis of the obtained results.

5.1. Quantitative comparison of the proposed approaches vs ResNet-18

Table 1 presents a comparison between the proposed strategies and ResNet-18 alone, for the ImageNetHVZ dataset. This comparison involves accuracy, average precision (micro-averaged area under the precision-recall curve), AUROC (Area Under the Receiver Operating Characteristic Curve), AOPC, POMPOM (averaged over the entire test set), and total number of training epochs.

AOPC was proposed by Samek et al. [18] as:

$$AOPC = \frac{1}{L+1}\left\langle \sum_{k=0}^{L} f\left(x_{MoRF}^{(0)}\right) - f\left(x_{MoRF}^{(k)}\right) \right\rangle_{p(x)} \quad (6)$$

where L is the number of perturbation steps, $f(x)$ is the function value (i.e. the softmax probability for the positive class) and $\langle . \rangle_{p(x)}$ denotes the average over all the images of the test set. This value is obtained by first dividing the heatmaps produced by the method under assessment in a predefined grid; then, computing the average pixel values of each grid patch and sorting these patches in descending order. Afterwards, each patch in the original image is perturbed, starting from the patch with higher heatmap relevance, adding the next most relevant patches at each step and computing $f(x)$. The intuition is that, by ranking the patches by relevance, $f(x)$ will suffer a steep decrease when the first patches are perturbed, as these were the ones that most influenced the classification outcome.

Finally, AOPC corresponds to the cumulative area over the MoRF curve at each step, and the values shown in Table 1 correspond to the AOPC value at the end of the perturbation process (the higher the better). Fig. 4(a) and (b) illustrate this process, and will be further discussed in Section 5.3. For a more in-depth understanding of this perturbation process we refer the reader to Figure 20 of [14].

We also propose POMPOM as a metric to compare between the influence of the proposed unsupervised and hybrid approaches. For a single instance, it is defined as the number of meaningful pixels outside the region of interest in relation to its total number of pixels, as presented in Eq. (7):

$$POMPOM = \left| \frac{\sum_{i,j}[(1 - z_{i,j})\hat{z}_{i,j}] > \epsilon}{\sum_{i,j}[(1 - z_{i,j})] > \epsilon + \epsilon} \right| \quad (7)$$

where $\hat{z}_{i,j}$ ($z_{i,j}$) is the value of pixel (i, j) of feature map $\hat{z}$ (mask z), and ϵ is a small factor ($\sim 1e{-}6$) to avoid divisions by 0. Only meaningful pixels are considered, i.e pixel values higher than ϵ. Therefore, POMPOM admits values in [0, 1] (the lower the value the better), going from black to white images, respectively.

We can conclude that the inclusion of the explainer module does not significantly decrease classification performance, although a decrease is tolerable to a certain extent, considering the accuracy-explainability trade-off [8]. Naturally, the joint architecture needs more training epochs, as a result of the added complexity. As expected, the POMPOM is lower for the hybrid approach, since it actively encourages explanations not to focus outside the region of interest. Finally, AOPC is slightly higher for the unsupervised approach.

Similar conclusions can be drawn from Table 2, which refers to the NIH-NCI Cervical Cancer dataset. In this case, POMPOM suffers a steeper decrease when comparing the unsupervised and hybrid approaches, as a significant area of the images is irrelevant (cervix walls and speculum).

5.2. Qualitative comparison of explanation quality

Fig. 3 depicts a comparison between some of the state-of-the-art explainability methods and the proposed approaches on the ImageNetHVZ dataset.[4] We also include a Canny edge detector and an Otsu segmentation for comparison.[5] These methods were tested

[2] When no histology is done, the patient’s cervix was considered healthy and did not require any further testing.

[3] https://github.com/albumentations-team/albumentations.git.

[4] For the same results on the NIH-NCI Cervical Cancer dataset we refer the reader to https://github.com/icrto/xML.git.

[5] An explanation can be coincident with a segmentation output, but this is not always the case, as discussed by Samek et al. [18]. Segmentation considers the whole object, while an explanation might be composed only by some parts of said object (for example, when distinguishing different breeds of dogs we are interested in properties like ear size and shape, nose, fur, etc.).

Table 1
Quantitative comparison between the proposed architecture and ResNet-18 for the ImageNetHVZ test set. Explanation quality (POMPOM and AOPC), as well as classification performance (accuracy, average precision and AUROC) are compared. Although the proposed approach takes longer to train, it produces meaningful explanations, while maintaining classification performance. Furthermore, the hybrid approach slightly outperforms the unsupervised approach in terms of POMPOM (the lower the better).

	Accuracy	Average precision	AUROC	AOPC	POMPOM	Epoch (per phase)
Classifier only	0.98	1.00	1.00	–	–	51
Unsupervised	0.95	1.00	1.00	0.310	0.190	10 + 10 + 28
Hybrid	0.95	0.99	0.99	0.302	0.132	10 + 10 + 53

Table 2
Quantitative comparison between the proposed architecture and ResNet-18 for the NIH-NCI Cervical Cancer test set. Explanation quality (POMPOM), as well as classification performance (accuracy, average precision and AUROC) are compared. Although the proposed approach takes longer to train, it produces meaningful explanations, while only slightly affecting classification performance. Furthermore, the hybrid approach outperforms the unsupervised approach in terms of POMPOM (the lower the better).

	Accuracy	Average precision	AUROC	POMPOM	Epoch (per phase)
Classifier only	0.88	0.71	0.82	–	19
Unsupervised	0.82	0.48	0.72	0.662	30 + 10 + 8
Hybrid	0.83	0.47	0.76	0.158	30 + 10 + 16

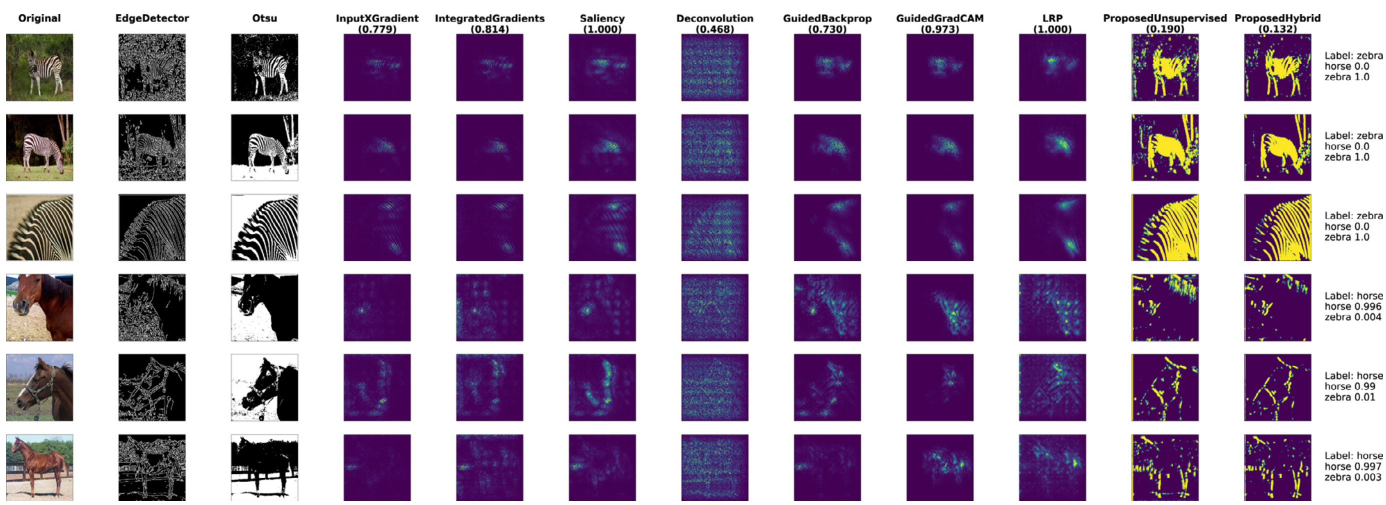


Fig. 3. Qualitative comparison of state-of-the-art methods and the proposed approaches on the ImageNetHVZ dataset. Each row corresponds to a different image, while each column depicts the output of several methods, including a Canny edge detector and an Otsu binarisation. For each image the classification score is presented, and for each method average POMPOM values for the test set are included. The colourmap ranges from purple to yellow (between [0, 1]) and absolute pixel values were considered for all methods for a fair comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

using the Captum library available for Pytorch,[6] except LRP-$\alpha_1\beta_0$.[7] Each explainability method is accompanied by the respective POMPOM value, and, for each image, we also compute the class confidence score.

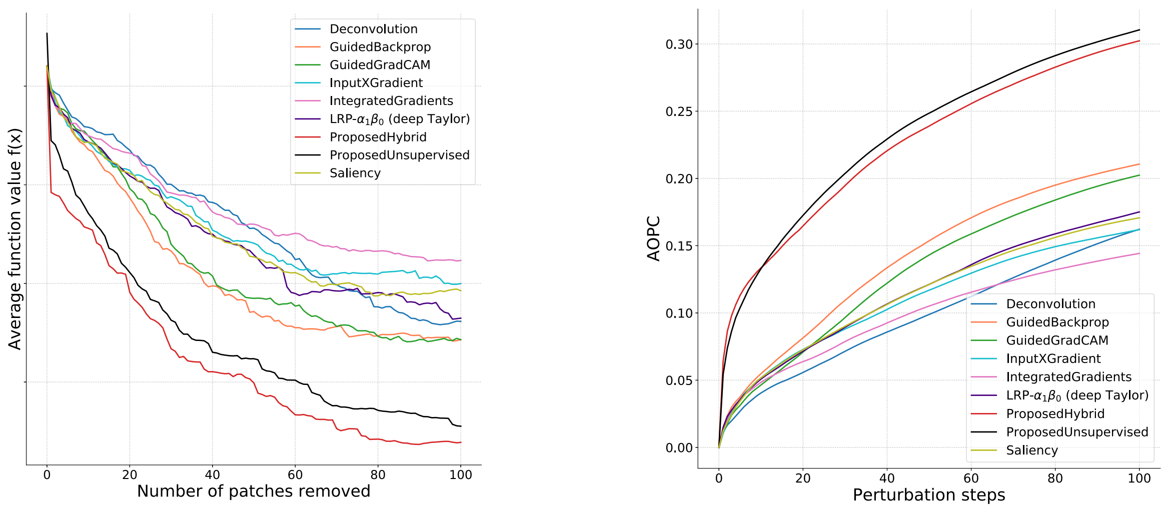
The majority of the state-of-the art methods produce visually meaningful explanations, image- and class-dependent. However, Deconvolution [24] produces much noisier explanations that resemble filter visualisations, only taking into account properties of individual images, fruit of the unpooling nature of this method [18].

The proposed approaches' explanations are also image- and class-dependent, while still being able to highlight in a clearer way the relevant image regions. Keeping in mind that the dataset is only composed by images of zebras and horses, it is clear that one of the main properties that distinguishes both animals is the texture of their fur, which is clearly highlighted by the proposed approaches. It is also worth noting that horses are identifiable via their riding equipment (5th row), and their legs and hooves (6th row). Moreover, the fact that explanations are focused on texture aligns with recent research by Geirhos et al. [6], in which the authors validated that ImageNet trained CNNs were biased towards texture. Finally, as expected, explanations produced by the hybrid approach present less active pixels outside the region of interest when compared to the ones produced by the unsupervised approach.

5.3. Quantitative comparison of explanation quality

Fig. 4(a) and (b) show quantitative results obtained on the ImageNetHVZ dataset using the perturbation process described in Section 5.1, proposed by Samek et al. [18] and Montavon et al. [14]. Both plots were obtained by perturbing patches of size 16 × 16 for 100 steps, through Gaussian Blurring. We tested several patch sizes to assess the impact of the perturbation in relation to kernel size, and other perturbation strategies (gray, white, black and random uniform) to guarantee independence from the model. These results

[6] https://captum.ai.
[7] This method was tested using the implementation available at https://github.com/myc159/Deep-Taylor-Decomposition.



(a) (b)

Fig. 4. Quantitative results of the ImageNetHVZ dataset, in terms of Average Function Value (a) and AOPC (b). These were obtained by iteratively perturbing image patches, from the most to the least relevant patches, as given by each of the methods. At each step, the classification outcome is computed, as well as the differences between the classification outcome of the unperturbed and perturbed images. The more relevant the patch, the greater the impact on the classification performance. These plots correspond to a 16 × 16 patch size and Gaussian Blur as perturbation strategy (other patch sizes and perturbation strategies available at https://github.com/icrto/xML.git). The proposed approaches clearly outperform state-of-the art methods (the higher the AOPC value the better).(For better analysis of these plots, the reader is referred to the web version of this article.)

can be found on our GitHub repository,[8] where the implementation of the proposed architecture is also publicly available.

As expected, the average function value decreases and AOPC increases with the number of patches removed (by most to least relevant). Furthermore, the hybrid approach slightly outperforms the unsupervised approach, and both outperform state-of-the-art methods. We also verified that, similarly to what was described by Samek et al. [18], with a patch size much larger than the kernel size (16 × 16 and 32 × 32), AOPC scores are higher, as we are obtaining a region score closer to the filter score, which means that perturbing that region largely impacts the filter response and, consequently, lowers the classification performance. Conversely, perturbing smaller regions (4 × 4 and 8 × 8) has less impact on the filter response, leading to smaller AOPC values.

6. Conclusions

Methods that allow AI systems to be understandable and trusted by humans are essential in many areas of application, especially if they are able to generate these explanations in an unsupervised fashion.

To tackle this problem we propose an in-model explainer + classifier joint approach to produce decisions and explanations using CNNs without the need to supervise the explanation component. This novel architecture, along with its three-phase training process and loss function (both unsupervised and hybrid approaches), allows for the classifier to be trained with the produced explanations, thus only focusing on relevant parts of the input image. Similarly, the explainer is trained together with the classifier, thus learning to find the relevant image regions for the classification. This architecture can be used with any classifier, provided the corresponding connections are made and the classifier is retrained within this new scenario.

The results obtained on both datasets show that this architecture is able to produce image- and class-dependent meaningful visual explanations, as well as decisions, while not degrading classification accuracy. From the quantitative assessment performed through an iterative perturbation process of the most relevant image regions, we also concluded that the proposed approach outperformed several state-of-the-art methods in terms of AOPC. To further compare both proposed approaches, we introduce a new metric, POMPOM, which measures the percentage of pixels outside the annotation mask. We also concluded that the hybrid strategy outperforms the unsupervised alternative, both in terms of AOPC and POMPOM.

Furthermore, the proposed approach can be improved in different research directions: design of different loss functions that include population level characteristics (promotion of explanation clusters similar to the decision clusters), and extension of the architecture to include multimodal inputs and outputs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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